Unit 6, Lesson 9: Exploring Multi-Step Proportion Problems

Benchmarks

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

MA.7.AR.4.5 Solve real-world problems involving proportional relationships.

Learning Target

☐ I can solve multi-step problems involving proportions.

6.9.1 Warm-Up: Baking with Ratios

Chef and author Michael Ruhlman makes baking easier by using simple mathematical ratios. He uses the following chart where each number represents the number of parts of that ingredient needed for the base recipe.

	Flour	Liquid	Egg	Butter	Sugar
Bread	5	3			
Quick Bread Muffins	2	2	1	1	1 *(if sweet)*
Cookies	3			2	1

1. Michael Ruhlman believes, "If you know ratios, it's like knowing a thousand recipes." What do you think he meant by this?

2. The ingredients for a homemade bread recipe are shown to the right. Do the ratios in the recipe match the ratios Michael recommended for bread? Explain your reasoning.

- ☐ 2 cups warm water
- ☐ $\frac{1}{2}$ cup white sugar
- ☐ $1\frac{1}{2}$ tablespoons yeast
- ☐ $1\frac{1}{2}$ teaspoons salt
- ☐ $\frac{1}{4}$ cup vegetable oil
- ☐ 5 – 6 cups flour

6.9.2 Guided Instruction: Thinking Through Multiple Steps

Solving real-world problems often involves multiple steps. To answer one question, a problem solver may need to ask and answer other questions using the information given in the problem.

Consider the following situation.

Tahj needs to paint the four exterior walls of a large rectangular barn. The length of the barn is 80 feet (ft), the width is 50 feet, and the height is 30 feet. The paint costs $\$28$ per gallon (gal), and each gallon covers 420 square feet.

Tahj wants to know how much it will cost to buy enough paint to paint the barn.

1. Complete the questions that Tahj needs to answer before she can determine the cost. Then, find the answer to each question in the workspace provided.

Question	Workspace
What is the total ______ Tahj needs to paint, in square feet?	
How many ___________ of paint will be needed to cover this area?	
How much will it cost Tahj to buy enough paint to paint the barn?	

2. Place a star next to any questions you answered using a proportion or proportional thinking.

3. Studying mathematics includes making connections between concepts when solving problems. In addition to proportional thinking, what other math concept(s) did you use to solve this problem?

6.9.3 Collaboration: Thinking Through Multiple Steps

Sean lives in Florida and frequently travels for work. Each week, he is allotted 500 U.S. dollars for business-related expenses while he is out of town. He is going to Canada this week, so he went to the bank to exchange 500 U.S. dollars for Canadian dollars (CDN) at a rate of $\$1.32$ CDN per $\$1$ U.S. On the way home from the bank, Sean's boss called to say that he has to go to Mexico City now instead. Sean went back to the bank to exchange his Canadian dollars for Mexican pesos at a rate of 16.21 pesos per $\$1$ CDN.

Two students used different strategies to determine how many pesos Sean has for his business trip. Their partial work is shown. Work with your partner to complete each student's work.

Maurice's Strategy Using Unit Rates	$\$500\,US \cdot \dfrac{\$1.32\,CDN}{\$1\,US} \cdot \dfrac{\_\_\_\_\_\,pesos}{\$\_\_\_\_\,CDN} =$ $\_\_\_\_\_\_\_\_\_\_\,pesos$
Kiara's Strategy Using Proportions	$\dfrac{\$1.32\,CDN}{\$1\,US} = \dfrac{x}{\$500\,US}$ $\dfrac{\$1.32\,CDN}{\$1\,US} \cdot 500 = \dfrac{x}{\$500\,US} \cdot 500$ $\$660\,CDN = x$ 1 $\dfrac{\$\_\_\_\_\,CDN}{\_\_\_\_\_\,pesos} = \dfrac{\$\_\_\_\_\_\,CDN}{m\,pesos}$ $m = \_\_\_\_\_\_\_\_\_\_\,pesos$

1. Draw an asterisk (*) next to the part of each student's work that represents Sean converting his U.S. dollars to Canadian dollars.

2. Draw a smiley face (☺) next to the part of each student's work that was used to determine the number of pesos Sean has for his business trip.

3. Both students used proportional reasoning to solve the problem in multiple steps. Which method do you prefer? Explain.

6.9.4 Try It: Solving Multi-Step Problems with Proportions

1. Reina is making s'mores kits with a handmade thank-you card as party favors for her upcoming birthday party. She wants to make 9 kits, one for each party guest. It takes her about 15 minutes to assemble each kit and she doesn't have much time.

    a. Work with your partner to determine the question(s) that should be answered if Reina is worried about having enough time to make 9 kits.

    b. Determine if Reina will be able to make all of the kits in 2 hours. Show your work.

2. A store has two different brands of laundry detergent. Laundrette can do 60 loads of laundry and costs $\$10.25$. Peroxy does 24 loads of laundry and costs $\$4.56$.

    a. Complete the statement.

    If Claudia wants to compare which detergent will cost her less per load of laundry, she should find a

    - ○ proportion
    - ○ unit rate

    for each brand.

    b. Which laundry detergent costs less per load?

3. Denise makes trail mix by combining 3 parts granola, $1\frac{1}{2}$ parts dried fruit, and $\frac{1}{2}$ part chocolate. She plans to make 20 cups of trail mix for her family's upcoming camping trip. How much of each ingredient does she need?

6.9.5 Collaboration: Comparing Thinking

With your partner, find another pair of partners to make a group of four.

1. Discuss and compare your work on the Try It problems. Verify your answers and resolve any differences. If necessary, revise your work.

6.9.6 Wrap-Up: Growth Mindset

There is power in believing you can improve. Complete each statement based on your learning today.

1. Today, I am still unsure about …

2. I will try to build my confidence by …